

Jiaming Shan

jiamingshan@ucsb.edu | shanjiaming.github.io

Profile

Direct-entry Ph.D. student at UCSB, currently interning at Tencent Cloud WorkBuddy on post-training, reinforcement learning, and RSI; planning to master out with a master's degree in June 2027.

Education

University of California, Santa Barbara (UCSB) Sep 2024 - Present
Ph.D. in Computer Science Santa Barbara, CA

- Research focus: LLM efficiency, sparse and long-context attention, speculative decoding, and autonomous agents.
- Advisor: Prof. Xifeng Yan; coursework in the top 5% of the program.

Shanghai Jiao Tong University Sep 2020 - Jun 2024
B.S. in Computer Science, ACM Honors Class Shanghai, China

- Selected for an honors program for the top 5% of computer science students; core-course GPA: **3.98/4.3**; rank: **3/36**.
- 2023 National Scholarship; Zhiyuan Honorary Scholarship, 2020 - 2023.

Research Experience

University of California, Santa Barbara (UCSB) Sep 2024 - Present
Graduate Researcher - LLM Systems and Agents Santa Barbara, CA

- Co-first-authored **Learning When Not to Attend Globally (All-or-Here Attention, AHA)** (arXiv preprint): designed a token-level binary router that switches each attention head between global and sliding-window attention.
- Implemented inference-speedup paths and experiments in PyTorch, Triton, and FlashAttention; AHA replaced up to **88%** of full-attention operations without performance loss and complements sparse-attention systems such as DuoAttention.
- Built benchmark and environment abstractions for **Proactive Agent Research Environment**, including finite-state-machine app models, goal inference, and multi-app orchestration evaluation.
- Explored end-to-end fine-tuning for DFlash acceleration.

Massachusetts Institute of Technology (MIT) Aug 2023 - Jun 2024
Summer Research, CSAIL Boston, MA

- Implemented physically constrained agents and helper bots in ThreeDWorld and helped build an LLM-based cooperative-agent benchmark published at **NeurIPS 2024** (Datasets and Benchmarks Track).

MIT / Shanghai Jiao Tong University Feb 2023 - May 2023
Summer Research (Remote) Remote

- Co-developed a framework that converts environment states into text prompts for cooperation; contributed to VirtualHome and heuristic-baseline implementations and conducted user studies comparing LLM agents with classical planning methods for an **ICLR 2024** paper.

Internship Experience

Tencent Cloud WorkBuddy Sep 2026 - Present
Large Language Model Agent Research Intern

- Trained large language model agents from experience data; conducted research on RSI.

Shanghai Yuanwo Intelligent Technology Co., Ltd. (RWKV) Mar 2026 - Jun 2026
Agent Engineering Intern Remote

- Built document summarization and retrieval pipelines around local, high-concurrency RWKV deployments.
- Added tool-output summarization and key-information extraction to reduce downstream context usage; generated HTML result pages concurrently to improve readability.

Riema Labs Mar 2025 - Jul 2025
Research Intern

- Contributed to MathEye research on automated repair for LLM-generated Lean proofs, based on *From Sorry to QED: Refining LLM Math Proofs via Angelic Execution*; used angelic execution to abstract failed fragments, proof dependency graphs and MaxSAT to select key obligations, and automatically reconstructed verifiable proofs.

Selected Publications

- [Learning When Not to Attend Globally](#) (AHA). Xuan Luo*, Jiaming Shan*, Wesley Truong, Xifeng Yan. arXiv preprint. (* Equal contribution.)
- [Proactive Agent Research Environment: Simulating Active Users to Evaluate Proactive Assistants](#). Deepak Nathani, Cheng Zhang, Chang Huan, Jiaming Shan, et al. arXiv preprint.
- [Building Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge](#). NeurIPS 2024, Datasets and Benchmarks Track.
- [Building Cooperative Embodied Agents Modularly with Language Models](#). ICLR 2024.

Open-Source Projects and Agent Systems

Sentinel-Rosette (Feb 2025; core prototype in two weeks): [GitHub](#). **Motivation:** In the Racket/Rosette ecosystem, reliable AI coding assistance was largely unavailable, and even parentheses and semantics required manual correction. Independently built a runnable symbolic-execution system: implemented a symbolic EVM in Rosette, integrated the Foundry toolchain, routed compiled and executed Solidity exploit contracts and trace dumps into symbolic execution, and used MaxSMT to search attacker parameters and profit-optimal solutions. Supported Bancor, Oryn, Cover, and PAID attack workflows; cross-checked Foundry and symbolic-execution traces for consistency and provided a Dockerized environment.

Claude Link (Aug 2025; prototype in half a day): [GitHub](#). **Motivation:** Native subagents often behave like one-shot function calls and cannot preserve full context, tool state, and human intervention at the same time. Let Claude Code use Claude Code: through MCP, ran each complete Claude Code session in an independent tmux pane and provided agent creation, shutdown, refresh, text and screen capture, inbox, forced injection, and completion hooks. Supported parent-child and peer-to-peer collaboration, forming a persistent, observable, and controllable network of agents.

Research Auto-Research and Technical Writing: Turned scientific question decomposition, literature search and synthesis, experiment planning, and intermediate reasoning into reusable research workflows; documented practical lessons in [a Zhihu technical article](#).

TA Grading Agent: Exposed Gradescope grading workflows as MCP tools and agent workflows for batch grading; used dual-pass grading and human verification queues to reduce TA workload by approximately **70%**.

Thought-to-Blog Agent: Built a pipeline that turns high-quality AI chat records into publishable blog drafts, with privacy and value screening plus working, final, and published versions.

Autonomous Investment Research: Built investment-research automation connected to moomoo and local agents, generating recommendations from portfolio context, supply-chain research, market conditions, tax and cash-flow constraints, and risk gates while maintaining a simulated-trading system.

Personal Agent Framework: Built and daily used long-running personal-agent workflows integrating memory, compaction, MCP tool routing, and multi-step task execution.

Honors and Awards

- 2023 National Scholarship Award.
- 2020 - 2023 Zhiyuan Honorary Scholarship, Shanghai Jiao Tong University (top 2%).
- 2021 Interdisciplinary Contest in Modeling, Honorable Mention.
- 2021 National Undergraduate Mathematical Contest in Modeling, Provincial Third Prize.
- 2021 International Physics Competition for University Students, Silver Medal.

Technical Skills

Programming Languages:	Python, C++, Racket/Rosette, Solidity
AI / Systems:	PyTorch, Hugging Face, Triton, FlashAttention, vLLM, Weights & Biases
Agent Workflows:	MCP, Claude Code, Codex, OpenClaw, Cursor, Tmux
Tools / Deployment:	Git, Docker

Links

[Personal Website](#) [Google Scholar](#) [GitHub](#) [Zhihu](#) [LinkedIn](#)